

Concurrent Semantics: Total Store Ordering (TSO)

Weak Memory Models (WMM):

- Unlike SC, the instructions of a thread may be **reordered**, i.e. they may execute **out of order**
- May observe **weak behaviours** (anomalies) not observable under SC.

TSO (Total Store Ordering)

- Allows the following for **adjacent** instructions (in program order):
 1. **write-read reordering on different locations**: a later read on y can be reordered before an earlier write on x when $x \neq y$.
 2. **write-read reordering on same location with write forwarding**: a later read on x can be reordered before an earlier write on x provided that the value of the write is forwarded to (inlined in) the read.
- 1 leads to **weak SB (store buffering)**; 2 alone **does not cause weak behaviours**.
- No other reordering or change is allowed
- i.e. TSO = SC + write-read reordering on different locations + write-to-read on same location + write forwarding